# A LOCAL NOETHERIAN RING WITH GEOMETRICALLY NORMAL FORMAL FIBRES HAS UNIVERSALLY CATENARY HENSELIZATION AND IS ITSELF UNIVERSALLY CATENARY WHEN UNIBRANCH

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 32497-32509 (canonical tag 0C2F) and the counted proof is the proof environment at source lines 32511-32523, which immediately follows it and is its own proof.

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*Editorial note.* The three references are all to earlier results of this chapter, namely the two branch-count statements and the Ratliff criterion that is itself the neighbouring published item; the kinds recorded are the type words appearing in the labels themselves.

*Editorial note.* The preamble span runs to line 90, the current end of the retained preamble, which is exactly what makes the reference environment a comment; the excerpt serves only as preamble, never as a span coordinate.

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*Allowed standard prerequisites (editorial).*

- For a Noetherian local ring with geometrically normal formal fibres the number of branches of the ring and of its completion agree. invoked at proof lines 32511-32513. Feeds the branch count into the criterion comparing branches with the completion, which is what produces the uniqueness of the minimal prime.
- If the number of branches of a local ring and its completion agree, then the completion of the henselization modulo a minimal prime has a unique minimal prime. invoked at proof lines 32513-32517. Gives the formal catenarity of the henselization by definition of that notion.
- A formally catenary Noetherian local ring is universally catenary, and a unibranch ring has henselization with a unique minimal prime. invoked at proof lines 32517-32522. Turns formal catenarity into universal catenarity in the first case, and supplies the uniqueness of the minimal prime that starts the second case.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0C2E: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32231–32231. The reference is applied at proof lines 32511-32512 to compare the numbers of branches of the ring and its completion; the use made of the comparison is inside the counted span.

*Source reference.* Original lemma 0C29: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31964–31964. The reference is applied at proof lines 32513-32515 to obtain the uniqueness of the minimal prime of the completion of the henselization; the conclusions drawn from it are inside the counted span.

*Source reference.* Original proposition 0AW6: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32484–32484. The reference is invoked twice, at proof lines 32517-32519 and again at proof lines 32522-32523, to pass from formal to universal catenarity; both uses are the last step of their respective case and the preparation is inside the counted span.

**Lemma 0.2.** *Let $(A, \mathfrak{m})$ be a Noetherian local ring with geometrically normal formal fibres. Then*

(1) *$A^h$ is universally catenary, and*
(2) *if $A$ is unibranch (for example normal), then $A$ is universally catenary.*

*Proof.* By Lemma 0C2E the number of branches of $A$ and $A^\wedge$ are the same, hence Lemma 0C29 applies. Then for any minimal prime $\mathfrak{q} \subset A^h$ we see that $A^\wedge/\mathfrak{q}A^\wedge$ has a unique minimal prime. Thus $A^h$ is formally catenary (by definition) and hence universally catenary by Proposition 0AW6. If $A$ is unibranch, then $A^h$ has a unique minimal prime, hence $A^\wedge$ has a unique minimal prime, hence $A$ is formally catenary and we conclude in the same way. □